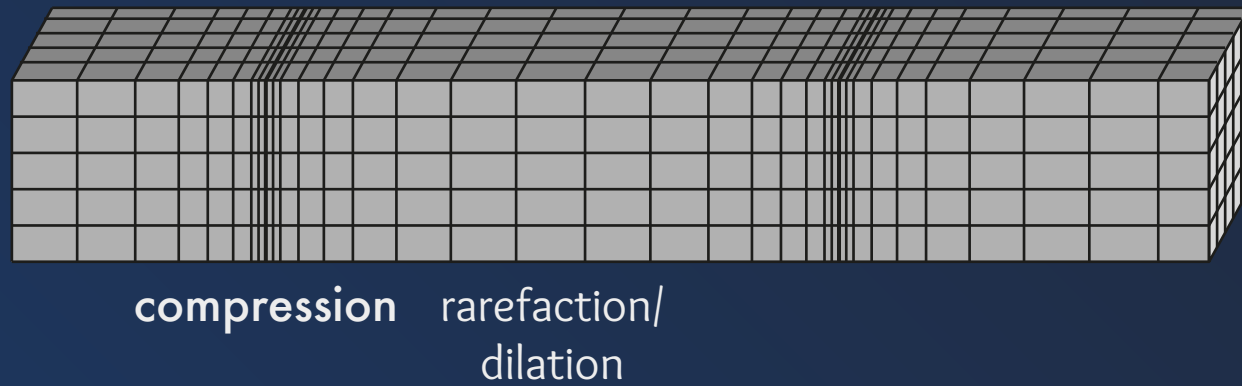


# Seismic (Body) Waves

## Pressure P-wave



propagation  
→  
direction

particle  
←→  
motion

Body waves travel through the interior

P-waves:

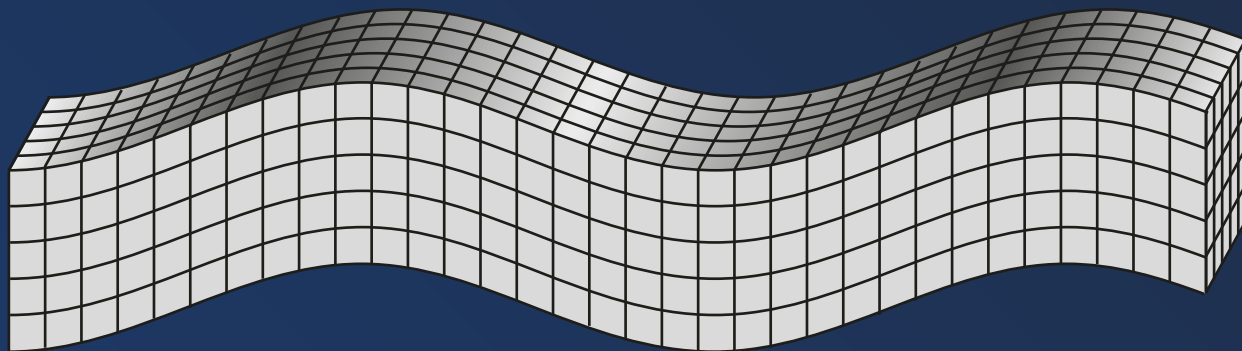
push-pull motion

fast, typically  $6-8 \text{ km s}^{-1}$  in crystalline rocks

faster in dense rocks

travels through solids and fluids (including air)

## Shear S-wave



propagation  
→  
direction

SV  
↑  
SH  
↗  
particle  
motion

S-waves:

side to side (SH) or up-down (SV) motion

slower than P-waves, typically  $3.5-5 \text{ km s}^{-1}$  in crystalline rocks

travels only in solids